

知识总结-导数与微分

1.可导定义:	最常用形式: $f'(x_0)=\lim_{\Delta x\rightarrow 0}\frac{f(x_0+\Delta x)-f(x_0)}{\Delta x}$ 单侧导数:$\left\{\begin{array}{l} \text{左导数: } f'_-(x_0)=\lim_{\Delta x\rightarrow 0^-}\frac{f(x_0+\Delta x)-f(x_0)}{\Delta x} \\ \text{右导数: } f'_+(x_0)=\lim_{\Delta x\rightarrow 0^+}\frac{f(x_0+\Delta x)-f(x_0)}{\Delta x} \\ f'(x_0)\text{存在}\Leftrightarrow \text{左导数与右导数存在且相等} \end{array}\right.$ 另一种形式: $f'(x_0)=\lim_{x\rightarrow x_0}\frac{f(x)-f(x_0)}{x-x_0},(x=x_0+\Delta x)$ 示例: $\left\{\begin{array}{l} \lim_{h\rightarrow 0}\frac{f(\cos h-1)}{h^2} \\ \lim_{h\rightarrow 0}\frac{f(h)-f(-h)}{h} \end{array}\right.$存在$\Rightarrow$连续函数$f(x)$在$x=0$处可导 可微$\left\{\begin{array}{l} \text{定义: }\Delta y=A\cdot\Delta x+o(\Delta x)=dy+o(\Delta x)=f'(x_0)\Delta x+o(\Delta x) \\ \text{考法: }\Delta y-dy\text{是}\Delta x\text{的高阶无穷小} \end{array}\right.$ 二阶导数定义: $f''(x_0)=\lim_{\Delta x\rightarrow 0}\frac{f'(x_0+\Delta x)-f'(x_0)}{\Delta x}$ 导数存在与连续性: 若$f^{(n)}(x_0)$存在, 则在x_0的某个领域内有$f^{(n-1)}(x)$且$f^{(n-1)}(x)$在x_0处连续
2.各类函数求导	1.对数求导法:$\left\{\begin{array}{l} \text{幂指函数或多项乘、除、乘方、开方的函数, 可先取对数再求导} \\ \text{例: 幂指函数}\left\{\begin{array}{l} u(x)^{v(x)}=e^{v(x)\ln u(x)} \\ \left[u(x)^{v(x)}\right]'=\left[e^{v(x)\ln u(x)}\right]' \end{array}\right.\text{示例: }y=x^x,y'=x^x(1+\ln x) \end{array}\right.$ 2.隐函数求导$\left\{\begin{array}{l} \text{①方程两边对}x\text{求导, 要记住}y\text{是}x\text{的函数} \\ \text{②隐函数存在定理: }\frac{dy}{dx}=-\frac{F'_x(x,y)}{F'_y(x,y)}(F(x,y)=0) \\ \text{③微分形式不变性} \end{array}\right.$ 3.参数方程求导$\left\{\begin{array}{l} \left\{\begin{array}{l} x=x(t) \\ y=y(t) \end{array}\right.\Rightarrow\left\{\begin{array}{l} y'_x=\frac{y'_t}{x'_t} \\ y''_{xx}=\frac{(y'_t)_t}{x'_t}=\frac{x'_t y''_t-y''_t x'_t}{(x'_t)^3} \end{array}\right. \end{array}\right.$ 4.反函数求导$\left\{\begin{array}{l} \frac{dx}{dy}=\frac{1}{\frac{dy}{dx}}=\frac{1}{y'(x)} \\ x''(y)=\frac{d}{dy}\left(\frac{1}{y'(x)}\right)=\frac{-y''}{[y'(x)]^3} \end{array}\right.$ 5.复合函数求导$\left[f(g(x))\right]'=f'(g(x))\cdot g'(x)$
3.高阶导	基本公式:$\left\{\begin{array}{l} (a^x)^{(n)}=a^x\ln^n a \quad (a>0); (e^{ax+b})^{(n)}=a^n e^{ax+b} \\ (\sin(kx+b))^{(n)}=k^n\sin(kx+b+n\cdot\frac{\pi}{2}); (\cos(kx+b))^{(n)}=k^n\cos(kx+b+n\cdot\frac{\pi}{2}) \\ (x^m)^{(n)}=m(m-1)\cdots(m-n+1)x^{m-n}(n\leq m) \\ (\ln(ax+b))^{(n)}=(-1)^{(n-1)}a^n\frac{(n-1)!}{(ax+b)^n} \\ \left(\frac{1}{ax+b}\right)^{(n)}=(-1)^na^n\frac{n!}{(ax+b)^{n+1}} \end{array}\right.$ 方法$\left\{\begin{array}{l} \text{核心}\left\{\begin{array}{l} \text{莱布尼茨: } (uv)^{(n)}=C_n^0u^{(n)}v^{(0)}+C_n^1u^{(n-1)}v^{(1)}+\cdots+C_n^nu^{(0)}v^{(n)} \\ \text{一: } f(x)=f(x_0)+f'(x_0)(x-x_0)+\frac{1}{2!}f''(x_0)(x-x_0)^2+\cdots+\frac{f^{(n)}(x_0)}{n!}(x-x_0)^n+R_n(x) \\ \text{二: } f(x)=a_0+a_1(x-x_0)+a_2(x-x_0)^2+\cdots+a_n(x-x_0)^n+R_n(x) \\ \text{三: }\frac{f^{(n)}(x_0)}{n!}=a_n\Rightarrow f^{(n)}(x_0)=a_n\cdot n! \end{array}\right. \\ \text{归纳法: 逐次求导, 探索规律. 示例: }\left(\frac{1}{x+1}\right)^{(n)}=(-1)^n\frac{n!}{(x+1)^{n+1}} \end{array}\right.$
4.单调性与极值	1.单调性定义: 任意$x_1<x_2$,恒有$f(x_1)<f(x_2)$或$f(x_1)>f(x_2)$ $\left\{\begin{array}{l} \text{若}f(x)\text{在}(a,b)\text{内}f'(x)\geq 0(\text{或}\leq 0), \text{且等号仅在有限个点处成立,} \\ \text{则函数}f(x)\text{在}(a,b)\text{上严格单调增加 (或减少)} \end{array}\right.$ 这说明: 严格单调增加不一定导数>0.例: $y=x^3$在$\mathbf{R}$上严格单调增加 3.极值定义: 在x_0的去心领域内$f(x)>f(x_0)$(或$f(x)<f(x_0)$)恒成立 注意: 极值点处不一定可导, 也不一定连续, 间断点可能是极值点 4.必要条件: 若函数$f(x)$在x_0处可导, 且在x_0处取极值, 则$f'(x_0)=0$. 5.第一充分条件: 若当x经过x_0时, $f'(x)\left\{\begin{array}{l} \text{由“+”变“-”, 则}f(x_0)\text{为极大值} \\ \text{由“-”变“+”, 则}f(x_0)\text{为极小值} \end{array}\right.$ 6.第二充分条件:$\left\{\begin{array}{l} \text{设}f''(x_0)\neq 0, \text{ 且}f'(x_0)=0\left\{\begin{array}{l} f''(x_0)<0\text{时, }f(x_0)\text{为极大值} \\ f''(x_0)>0\text{时, }f(x_0)\text{为极小值} \end{array}\right. \\ \text{注意: 不是必要条件. 如: 若函数}f(x)\text{二阶可导且}x_0\text{处取极大值, 则}f''(x_0)\leq 0 \end{array}\right.$ 7.求极值步骤:$\left\{\begin{array}{l} \text{①求驻点 (或不可导的点): }f'(x)=0\Rightarrow x=x_0 \\ \text{②判别是否为极值点} \end{array}\right.$
5.凹凸性与拐点	1.凹凸性定义:任意不同两点x_1,x_2,恒有$f(\frac{x_1+x_2}{2})<\frac{f(x_1)+f(x_2)}{2}$或$f(\frac{x_1+x_2}{2})>\frac{f(x_1)+f(x_2)}{2}$ 2.凹凸性判别: 若在I上$f''(x)<0$或$f''(x)>0$, 则$f(x)$在I上是凸的 (或凹的) . 3.拐点判别定理1:当x变动经过x_0时, $f''(x)$变号, 则$(x_0,f(x_0))$为拐点. 4.拐点判别定理2: $f''(x_0)=0, f'''(x_0)\neq 0$,则$(x_0,f(x_0))$为拐点. 5.求拐点的方法:$\left\{\begin{array}{l} \text{1.求}f',f'' \\ \text{2.令}f''=0(\text{或二阶导不存在的点})\Rightarrow x=x_0 \\ \text{3.按照判别定理判断是否为拐点} \end{array}\right.$

6.极值拐点:	判别: $f'(x_0)=f''(x_0)=\cdots=f^{(n-1)}(x_0)=0, f^{(n)}(x_0)\neq 0, \left\{\begin{array}{l} \text{极值点, }n\text{为偶数} \\ \text{拐点, }n\text{为奇数} \end{array}\right.$ 同时: 曲线的可导点不可同时为极值点和拐点; 曲线的不可导点可同时
7.函数示意图绘制:	1.求出单调区间 2.考虑极值与渐近线(无需求解表达式) 3.部分试题可用凹凸性快速画出
8.曲线的渐近线	种类$\left\{\begin{array}{l} \text{垂直渐近线: }\lim_{x\rightarrow x_0}y=\infty(\text{或}\lim_{x\rightarrow x_0}y=-\infty) \\ \text{水平渐近线: }\lim_{x\rightarrow+\infty}y=c(\text{或}\lim_{x\rightarrow-\infty}y=c) \\ \text{斜渐近线: }\lim_{x\rightarrow+\infty}\frac{y}{x}=k(k\neq 0); \lim_{x\rightarrow+\infty}(y-kx)=b(b\neq 0)\text{同理} \end{array}\right.$ 注意$\left\{\begin{array}{l} \text{求解时按照①垂直②水平③斜渐线的顺序} \\ \text{在同一侧水平与斜渐近线不得兼得} \\ \text{正负无穷有时候需要区分, 分为左右两侧分别去求} \\ \text{求斜渐近线时, 仅求出}k\text{不能确定有斜渐近线} \\ \text{若}\lim_{x\rightarrow+\infty}(y-kx-b)=0(x\rightarrow-\infty\text{同理}),\text{则有渐近线}y=kx+b \end{array}\right.$ 示例:$\left\{\begin{array}{l} \text{曲线}y=x+\sin x\text{无渐近线} \\ \text{曲线}y=\frac{1}{x}+\ln(1+e^x)\text{有渐近线}x=0,y=0,y=x \end{array}\right.$
9.求函数值域	锁定区域$\left\{\begin{array}{l} \text{周期函数: 只需求一个周期上的值域} \\ \text{奇偶函数: 只需求}x\geq 0(x\leq 0)\text{时的值域} \\ \text{普通函数: 求全部定义域上的值域} \end{array}\right.$ 值域求解: 比较驻点, 不可导点, 端点函数值的大小, 得出最大(小)值
10.曲率(数三不要求)	$\left\{\begin{array}{l} y=f(x)\text{时, }k=\frac{ y'' }{(1+y'^2)^{\frac{3}{2}}} \\ x=\varphi(t)\text{时, }k=\frac{ \varphi'(t)\psi''(t)-\varphi''(t)\psi'(t) }{[\varphi'^2(t)+\psi'^2(t)]^{\frac{3}{2}}} \end{array}\right.$ 曲率半径: $\rho=\frac{1}{k}$
11.导数经济意义(仅数三)	基本概念:$\left\{\begin{array}{l} \text{变量的含义: }Q\text{需求; }P\text{价格; }R\text{收益; }C\text{成本} \\ \text{关系式}\left\{\begin{array}{l} \text{收益}R=P\cdot Q \\ \text{利润}R=R(Q)-C(Q) \end{array}\right. \end{array}\right.$ 边际函数$\left\{\begin{array}{l} \text{设函数}f(x)\text{在}x\text{处可导, 则导数}f'(x)\text{为}f(x)\text{的边际函数} \\ \text{边际成本: 成本函数}C(Q)\text{的导数}C'(Q)\text{叫做边际成本} \\ \text{边际收益: 收益函数}R(Q)\text{的导数}R'(Q)\text{叫做边际收入} \\ \text{边际利润: 利润函数}L(Q)\text{的导数}L'(Q)\text{叫做边际利润} \end{array}\right.$ 需求价格弹性$E_d=\frac{dQ}{dP}\cdot\frac{P}{Q}$